

EPFD-Aware Beam Reassociation for Service Recovery in Multi-Beam LEO Satellite Systems under NGSO–GSO Coexistence

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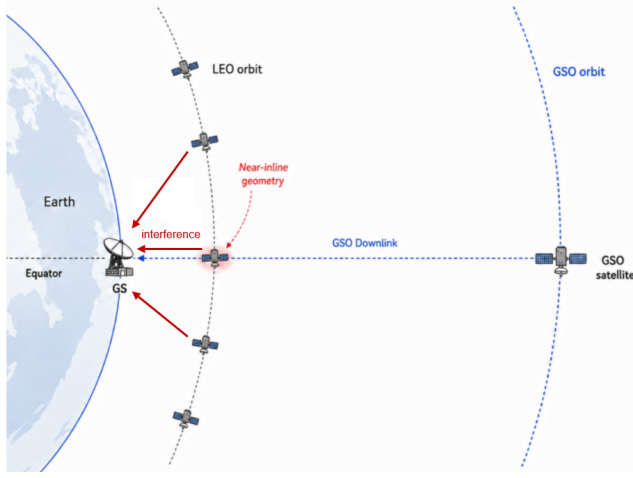


Fig. 1. NGSO–GSO coexistence scenario and near-inline interference geometry between a LEO satellite, a GSO ground station, and the target GSO satellite.

I. INTRODUCTION

In recent years, non-geostationary satellite orbit (NGSO) systems, especially low Earth orbit (LEO) constellations, have become an important component of broadband satellite communications, non-terrestrial networks (NTN), and future 6G architectures [1], [2], [7], [8]. Unlike geostationary satellite orbit (GSO) systems, which are located above the Earth's equator and appear fixed with respect to the ground, LEO satellites move rapidly along their orbits and can provide lower latency and more flexible coverage. However, when NGSO downlink transmissions operate in the same or adjacent frequency bands as existing GSO systems, they may cause co-channel interference to GSO ground stations. In particular, when a LEO satellite passes close to the line-of-sight direction between a GSO ground station and its target GSO satellite, a near-inline geometry may occur, leading to a higher interference risk, as illustrated in Fig. 1.

To protect existing GSO systems from interference generated by NGSO systems, the International Telecommunication Union (ITU) specifies interference limits for NGSO systems in Article 22 of the Radio Regulations [9]. Among these limits, equivalent power flux-density (EPFD) is used to evaluate the aggregate interference generated by NGSO transmissions toward GSO systems. Since EPFD depends on satellite geometry, transmit power, antenna gain, propagation loss, and the off-axis angle observed at the GSO ground station, its value

changes over time as LEO satellites move along their orbits. Therefore, an NGSO system must ensure that the aggregate EPFD received at the GSO ground station remains below the applicable EPFD limit.

To satisfy the EPFD constraint, NGSO systems may adopt different interference mitigation strategies, such as transmit power control, satellite tilt or pitch adjustment, beam shutdown, and beam steering [17]–[19]. These approaches can reduce the interference observed at the GSO ground station by lowering the transmitted power toward the GSO direction, changing the satellite pointing geometry, or avoiding high-risk beam transmissions. However, such mitigation actions may also reduce the available service resources for LEO users. For example, reducing beam power can decrease the achievable downlink rate, while shutting down high-risk beams can directly interrupt the service of users originally covered by those beams. Therefore, NGSO–GSO coexistence should not only be evaluated from the perspective of EPFD compliance, but also from the perspective of service continuity and user satisfaction.

In a multi-beam LEO system, the service impact of EPFD mitigation becomes more critical during near-inline events. When a LEO satellite passes close to the GSO ground station and the GSO line-of-sight direction, its beams may contribute a large portion of the aggregate EPFD. To satisfy the EPFD limit, the system may need to shut down some high-risk beams on the dominant interfering satellite. Although beam shutdown is effective in reducing the EPFD contribution toward the GSO ground station, it also interrupts the service of users originally served by those closed beams. Therefore, the users under closed beams become the main service demand that must be recovered during the critical period.

This service recovery problem creates an opportunity for beam reassociation. LEO satellite motion and beam footprints are predictable, so the coverage relationship among neighboring satellites can be estimated in advance. In particular, same-orbit neighboring satellites may provide sufficient coverage overlap with the dominant interfering satellite during a critical period. Therefore, users affected by closed-beam shutdown may still be covered by neighboring LEO beams and can potentially be reassociated to these beams for service recovery. This observation motivates the use of neighboring satellites as helper satellites for EPFD-aware service recovery through beam reassociation.

Based on this observation, this work proposes EPFD-Aware Beam Reassociation for Service Recovery (EABR), a beam

reassociation framework for multi-beam LEO satellite systems under NGSO–GSO coexistence. The main idea is to exploit the predictable coverage overlap among neighboring LEO satellites. When high-risk beams are shut down to satisfy the EPFD limit, users originally served by those beams may still be covered by nearby helper beams. The proposed EABR framework therefore uses beam reassociation to transfer affected users from closed beams to feasible helper recovery beams, so that their service can be continuously supported during the critical period. In this way, EABR aims to improve user satisfaction while maintaining EPFD compliance.

The main contributions of this work are summarized as follows.

- We propose EPFD-Aware Beam Reassociation for Service Recovery (EABR), a beam reassociation framework for EPFD-constrained multi-beam LEO systems. EABR exploits predictable coverage overlap among neighboring LEO satellites to continuously support users affected by critical-beam shutdown.
- We design the core components of the proposed framework, including a time-slot beam role record, Safe-Beam Reassociation for Helper Users (SBR), released power allocation for helper recovery beams, and Helper-Beam Reassociation for Closed-Beam Users (HBR). The time-slot beam role record captures critical satellites, helper satellites, and overlap-based candidate beam relation graphs. SBR uses critical safe beams to reassociate selected helper users and release helper satellite transmit power. The released power is then allocated to helper recovery beams, while HBR reassociates closed-beam users to feasible helper recovery beams.
- We evaluate the proposed framework using an STK–MATLAB simulation framework [25], [26]. The results show that the proposed framework satisfies the EPFD constraint while improving average user satisfaction, enhancing service recovery capability, and reducing the proportion of low-satisfaction users compared with baseline methods.

II. RELATED WORK

Existing studies on NGSO–GSO coexistence have investigated several interference mitigation strategies for reducing NGSO downlink interference toward GSO systems. In this chapter, related works are reviewed in three categories: progressive pitch and beam shutdown, joint power and tilt control, and beamforming-based interference mitigation. The characteristics and limitations of these methods are discussed to clarify the motivation of this work.

A. Progressive Pitch and Beam Shutdown

Progressive pitch is one representative attitude-based interference avoidance strategy for NGSO–GSO coexistence. In this approach, the LEO satellite gradually adjusts its attitude when it approaches low-latitude regions, where its downlink direction may become close to the line-of-sight direction from the GSO ground station to the target GSO satellite. By changing the satellite pointing direction, the LEO downlink beam

can be shifted away from this line-of-sight direction, thereby increasing the angular separation observed at the GSO ground station. When attitude adjustment alone is insufficient to satisfy the interference constraint, some high-risk user beams may be shut down to further reduce interference.

Li et al. [19] studied the progressive pitch strategy of the OneWeb constellation. Their results show that progressive pitch can effectively reduce interference to GSO systems in many regions. However, the combination of satellite tilting and partial beam shutdown may also reduce coverage continuity and create service blind areas during certain periods. This indicates that interference avoidance based only on attitude adjustment and beam shutdown may still affect the service availability of LEO users.

B. Joint Power and Tilt Control

Another type of interference mitigation method combines transmit power control with satellite tilt control. Power control reduces the transmit power of LEO satellites when they approach high-risk regions near the GSO ground station, so that the interference received at the GSO ground station can be lowered. Compared with progressive pitch methods that follow a predefined attitude adjustment strategy, joint power and tilt control formulates the mitigation process as an optimization problem. The transmit power and tilt angle of LEO satellites are jointly adjusted to satisfy the EPFD constraint while improving user demand satisfaction.

Jalali et al. [18] proposed a joint power and tilt control method for NGSO–GSO interference mitigation. Their method optimizes LEO transmit power and satellite tilt angle under the aggregate EPFD constraint. Compared with direct beam shutdown, this approach provides a more flexible way to reduce interference and preserve user service. However, when the interference constraint is strict, the transmit power of critical satellites may still need to be significantly reduced, which can limit the achievable service quality of LEO users during critical periods.

C. Beamforming and Beam Steering

Beamforming-based methods provide another interference mitigation direction for NGSO satellite systems equipped with phased-array antennas. Instead of only reducing transmit power or adjusting satellite attitude, beamforming controls antenna weight vectors to steer signals toward intended LEO users and suppress emissions toward GSO ground stations. In this way, the NGSO system can improve its downlink transmission while reducing co-channel interference to the GSO system.

Jalali et al. [20] investigated downlink beamforming strategies for interference-aware NGSO satellite systems. They proposed an aggregate interference-constrained beamforming method to limit the interference received at GSO ground stations while satisfying the service requirements of LEO users. This approach shows that beamforming can be an effective interference mitigation technique for advanced NGSO payloads. However, beamforming generally requires phased-array antennas, channel state information or average channel state information, and higher computational complexity.

Since this work focuses on a fixed multi-beam LEO system, beamforming-based control is not considered in this thesis.

D. Summary and Research Gap

The above studies show that NGSO–GSO interference can be mitigated through satellite attitude adjustment, transmit power control, beam shutdown, or antenna-domain beamforming. These methods reduce the interference received at GSO ground stations from different perspectives. However, when mitigation actions reduce the available resources of LEO satellites, the service quality of LEO users may still be affected. In particular, users originally served by beams that are shut down during critical periods require additional service recovery support.

Existing studies have not addressed how neighboring LEO satellite beams can be used to recover the service of users affected by critical-beam shutdown under the GS EPFD constraint. This motivates the proposed EABR framework, which exploits predictable coverage overlap among neighboring LEO satellites for service recovery while maintaining EPFD compliance.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

1) Network Scenario: We consider a general multi-beam LEO satellite communication system under NGSO–GSO co-existence. Since the LEO downlink and the GSO system may operate in the same frequency band, downlink transmissions from visible LEO satellites can generate co-channel interference at a GSO ground station. Hereafter, this ground station is referred to as the GS. The GS communicates with its target GSO satellite, and the direction from the GS to the target GSO satellite is used as the reference direction for EPFD evaluation. When a visible LEO satellite covers the GS region and also passes close to this reference direction as observed from the GS, a near-inline NGSO–GSO geometry may occur, leading to a higher interference risk.

The system is modeled in discrete time slots. Let $\mathcal{T}$ denote the set of time slots considered in this work, and let $t \in \mathcal{T}$ denote a specific time slot. For each time slot t , the satellite geometry, beam footprints, user coverage, and EPFD contribution may change due to LEO satellite motion. Therefore, the system state is evaluated on a time-slot basis.

Let $\mathcal{S}(t)$ denote the set of visible LEO satellites at time slot t , and let $\mathcal{B}_s$ denote the set of beams on satellite s . Each satellite employs a fixed multi-beam downlink layout, where each beam has a predefined pointing direction relative to the satellite and forms a corresponding ground footprint. For beam b on satellite s , the ground footprint at time slot t is denoted by $\mathcal{F}_{s,b}(t)$. If user u lies within $\mathcal{F}_{s,b}(t)$, beam b on satellite s is regarded as a feasible serving candidate for that user.

Because neighboring LEO satellites and their beams may have overlapping ground footprints, a terrestrial user may be covered by multiple beams from visible LEO satellites in the same time slot. This overlap provides multiple feasible serving beams for user association and resource coordination.

In this work, these feasible beams are assumed to operate with compatible downlink frequency and polarization settings. Therefore, changing the serving beam of a user can be modeled as a serving-beam reassociation within the feasible coverage set. This assumption is consistent with LEO satellite handover and beam management concepts, where predictable satellite motion, overlapping beam coverage, and compatible radio resources enable service continuity when the serving beam or serving satellite changes [3]–[6].

2) User Association and Service Model: Let $\mathcal{U}(t)$ denote the set of terrestrial users considered at time slot t . Each user can be served by at most one beam in the same time slot. The user association is represented by a binary indicator $x_{u,s,b}(t)$, defined as

$$x_{u,s,b}(t) = \begin{cases} 1, & \text{if user } u \text{ is served by beam } b \text{ of satellite } s, \\ 0, & \text{otherwise,} \end{cases}$$

for all $u \in \mathcal{U}(t)$, $s \in \mathcal{S}(t)$, and $b \in \mathcal{B}_s$. A user can only be associated with a beam that covers its location. Therefore, if user u is not located within the footprint $\mathcal{F}_{s,b}(t)$, beam b of satellite s is not a feasible serving beam for this user.

For beam b of satellite s , let $N_{s,b}(t)$ denote the number of users served by this beam at time slot t . This value represents the beam loading and is used later in the downlink rate model, where users served by the same beam share the available downlink resource.

Each user u has a service demand, denoted by D_u . The demand represents the required downlink data rate of the user. In addition, different users may have different service priorities. Motivated by the priority concept in 3GPP QoS differentiation [15], each user u is assigned a priority class $\pi(u)$, and each priority class is associated with a priority weight $w_{\pi(u)}$. A higher priority weight indicates that the user has a higher importance in service recovery and resource allocation.

The user association state at time slot t is denoted by $\mathbf{X}(t)$, which contains all association indicators $x_{u,s,b}(t)$. The association decisions across all time slots are denoted by $\mathbf{X} = \{\mathbf{X}(t) \mid t \in \mathcal{T}\}$.

3) Beam Power Model: Each beam is assigned a transmit power in each time slot. Let $p_{s,b}(t)$ denote the transmit power of beam b on satellite s at time slot t . The beam power allocation state at time slot t is denoted by $\mathbf{P}(t)$, which contains all beam transmit powers $p_{s,b}(t)$. The beam power allocation decisions across all time slots are denoted by $\mathbf{P} = \{\mathbf{P}(t) \mid t \in \mathcal{T}\}$.

Under normal operation, each beam is assigned a predefined default transmit power. The default beam-power profile is configured according to beam geometry, antenna gain, and propagation loss. The purpose is to provide ground users in different beam footprints with similar received downlink power. This default setting represents the baseline power allocation before EPFD-aware power adjustment or beam shutdown is applied.

Each beam has a beam-level power upper bound, denoted by $P_{s,b}^{\max}$, and each satellite has a total transmit power budget, denoted by $P_s^{\max}$. During EPFD-aware operation, the transmit

power of a beam may be adjusted from its default value or set to zero. A beam is regarded as shut down when its transmit power is set to zero. The beam-level and satellite-level power limits are included later in the problem formulation as power constraints.

4) SINR Model: The downlink SINR of a user is modeled using a standard wireless and satellite communication formulation, where the received desired signal, LEO co-channel interference, and receiver noise are considered. If user u is served by beam b of satellite s at time slot t , its downlink SINR is given by

$$\text{SINR}_u(t) = \frac{S_u(t)}{I_u^{\text{intra}}(t) + I_u^{\text{inter}}(t) + N_u(t)}, \quad (1)$$

where $S_u(t)$ is the desired signal power, $I_u^{\text{intra}}(t)$ is the LEO intra-satellite co-channel interference, $I_u^{\text{inter}}(t)$ is the LEO inter-satellite co-channel interference, and $N_u(t)$ is the receiver noise power.

Desired Signal Power: The desired signal power follows the standard satellite link-budget relation. For user u served by beam b of satellite s , the received desired signal power is given by

$$S_u(t) = \frac{p_{s,b}(t) G_{s,b \rightarrow u}^{\text{tx}}(t) G_u^{\text{rx}}}{L_{s,u}(t)},$$

where $p_{s,b}(t)$ is the transmit power of beam b on satellite s , $G_{s,b \rightarrow u}^{\text{tx}}(t)$ is the transmit antenna gain of beam b toward user u , G_u^{rx} is the user terminal receive gain, and $L_{s,u}(t)$ is the total propagation loss between satellite s and user u .

The total propagation loss includes free-space path loss and atmospheric gaseous loss:

$$L_{s,u}(t) = L_{s,u}^{\text{fs}}(t) L_{s,u}^{\text{gas}}(t),$$

where $L_{s,u}^{\text{fs}}(t)$ is the free-space path loss and $L_{s,u}^{\text{gas}}(t)$ is the atmospheric gaseous loss. The free-space path loss is given by

$$L_{s,u}^{\text{fs}}(t) = \left(\frac{4\pi d_{s,u}(t)}{\lambda} \right)^2,$$

where $d_{s,u}(t)$ is the distance between satellite s and user u , and λ is the carrier wavelength.

LEO Intra-Satellite Interference: The LEO intra-satellite interference comes from co-channel beams on the same satellite. Let $\mathcal{B}_s^{\text{co}}(b)$ denote the set of beams on satellite s that use the same downlink frequency resource as the serving beam b , excluding beam b itself. The intra-satellite interference received by user u is given by

$$I_u^{\text{intra}}(t) = \sum_{b' \in \mathcal{B}_s^{\text{co}}(b)} \frac{p_{s,b'}(t) G_{s,b' \rightarrow u}^{\text{tx}}(t) G_u^{\text{rx}}}{L_{s,u}(t)}.$$

LEO Inter-Satellite Interference: The LEO inter-satellite interference comes from co-channel beams on other visible LEO satellites. For another visible satellite s' , let $\mathcal{B}_{s'}^{\text{co}}(b)$ denote the set of beams on satellite s' that use the same downlink frequency resource as the serving beam b . The inter-satellite interference received by user u is given by

$$I_u^{\text{inter}}(t) = \sum_{\substack{s' \in \mathcal{S}(t) \\ s' \neq s}} \sum_{b' \in \mathcal{B}_{s'}^{\text{co}}(b)} \frac{p_{s',b'}(t) G_{s',b' \rightarrow u}^{\text{tx}}(t) G_u^{\text{rx}}}{L_{s',u}(t)},$$

where $G_{s',b' \rightarrow u}^{\text{tx}}(t)$ is the transmit antenna gain of the interfering beam toward user u , and $L_{s',u}(t)$ is the total propagation loss between satellite s' and user u .

With this SINR model, the user SINR changes when the serving beam, beam transmit power, propagation loss, or LEO co-channel interference changes.

5) Capacity and User Satisfaction Model: After obtaining the user SINR, the achievable downlink capacity is modeled by the Shannon capacity formula, which is commonly used in wireless and satellite communication systems. Let B denote the downlink bandwidth of a beam. If user u is served by beam b of satellite s at time slot t , the achievable downlink capacity of user u is given by

$$C_u(t) = B \log_2(1 + \text{SINR}_u(t)).$$

When multiple users are served by the same beam, they share the downlink resource of that beam. Following the time-division resource sharing model commonly used in satellite downlink systems, the beam resource is assumed to be equally divided among the users served by the same beam. The actual service rate of user u is therefore

$$R_u(t) = \frac{C_u(t)}{N_{s,b}(t)}, \quad (2)$$

where $N_{s,b}(t)$ is the number of users served by beam b of satellite s at time slot t . This expression represents an equal-time sharing model, where users served by the same beam are allocated the same fraction of the beam resource in each time slot.

Let D_u denote the demand rate of user u . The satisfaction of user u is defined as

$$s_u(t) = \min\left(\frac{R_u(t)}{D_u}, 1\right), \quad (3)$$

where $s_u(t) \in [0, 1]$. If $s_u(t) = 1$, the demand of user u is fully met; if $0 < s_u(t) < 1$, user u receives partial service; if $s_u(t) = 0$, user u receives no effective service in the time slot. This satisfaction metric represents the fraction of user demand that is satisfied in each time slot.

6) Beam-Level EPFD Model: The EPFD calculation follows the ITU Radio Regulations and ITU-R S.1503 framework for NGSO–GSO coexistence [9], [10]. Since different beams on the same LEO satellite may have different pointing directions, antenna gains, and transmit powers toward the GS, the EPFD contribution is evaluated at the beam level and summed in the linear domain.

For EPFD calculation, let $P_{s,b}^{\text{ref}}(t)$ denote the radio-frequency (RF) transmit power of beam b on satellite s within the ITU reference bandwidth B_{ref} specified for the applicable EPFD limit [11], [12]. If the beam transmit power $p_{s,b}(t)$ is defined over a wider beam bandwidth, it is converted to the corresponding RF power within B_{ref} before applying the EPFD formula. The beam-level EPFD contribution toward the GS is given by

$$\text{EPFD}_{s,b}(t) = \frac{P_{s,b}^{\text{ref}}(t) G_{s,b \rightarrow g}^{\text{tx}}(t)}{4\pi d_{s,g}^2(t)} \cdot \frac{G_g^{\text{rx}}(\theta_{g \rightarrow s}(t))}{G_{g,\text{max}}^{\text{rx}}},$$

where $G_{s,b \rightarrow g}^{\text{tx}}(t)$ is the transmit antenna gain of beam b on satellite s in the direction of the GS, $d_{s,g}(t)$ is the distance

between satellite s and the GS, $G_g^{\text{rx}}(\theta_{g \rightarrow s}(t))$ is the receive antenna gain of the reference GS antenna in the direction of satellite s , $G_{g,\text{max}}^{\text{rx}}$ is the maximum receive antenna gain of the reference GS antenna, and $\theta_{g \rightarrow s}(t)$ is the off-axis angle between the GS antenna boresight and the direction of satellite s .

The transmit antenna gain $G_{s,b \rightarrow g}^{\text{tx}}(t)$ is evaluated from the beam antenna pattern according to the off-axis angle between the boresight of beam b and the direction of the GS. Since each beam has its own pointing direction, the EPFD contribution of each beam is computed individually. If a beam is shut down, its transmit power is zero and its EPFD contribution is also zero.

The aggregate EPFD at the GS is obtained by summing the beam-level EPFD contributions from all active beams of all visible LEO satellites:

$$\text{EPFD}_{\text{agg}}(t) = \sum_{s \in \mathcal{S}(t)} \sum_{b \in \mathcal{B}_s} \text{EPFD}_{s,b}(t).$$

The resulting aggregate EPFD is compared with the applicable EPFD limit in the problem formulation.

B. Problem Formulation

Based on the system model above, we formulate the user association and beam power allocation problem under user association, beam power, satellite power budget, and EPFD constraints. The decision variables are the user association sequence $\mathbf{X}$ and the beam power allocation sequence $\mathbf{P}$. Specifically, $\mathbf{X}(t) = \{x_{u,s,b}(t)\}$ denotes the user association state at time slot t , and $\mathbf{P}(t) = \{p_{s,b}(t)\}$ denotes the beam power allocation state at time slot t .

Objective function:

$$\max_{\mathbf{X}, \mathbf{P}} \sum_{t \in \mathcal{T}} \sum_{u \in \mathcal{U}(t)} w_{\pi(u)} s_u(t). \quad (4)$$

Constraints:

$$\sum_{s \in \mathcal{S}(t)} \sum_{b \in \mathcal{B}_s} x_{u,s,b}(t) \leq 1. \quad (5)$$

$$x_{u,s,b}(t) \in \{0, 1\}. \quad (6)$$

$$0 \leq p_{s,b}(t) \leq P_{s,b}^{\text{max}}. \quad (7)$$

$$\sum_{b \in \mathcal{B}_s} p_{s,b}(t) \leq P_s^{\text{max}}. \quad (8)$$

$$\text{EPFD}_{\text{agg}}(t) \leq \text{EPFD}_{\text{lim}}. \quad (9)$$

where (5) holds for all $u \in \mathcal{U}(t)$ and $t \in \mathcal{T}$; (6) holds for all $u \in \mathcal{U}(t)$, $s \in \mathcal{S}(t)$, $b \in \mathcal{B}_s$, and $t \in \mathcal{T}$; (7) holds for all $s \in \mathcal{S}(t)$, $b \in \mathcal{B}_s$, and $t \in \mathcal{T}$; (8) holds for all $s \in \mathcal{S}(t)$ and $t \in \mathcal{T}$; and (9) holds for all $t \in \mathcal{T}$.

Constraint (5) ensures that each user is served by at most one beam in each time slot. As described in the user association model, only beams that cover the user are considered feasible serving beams.

Constraint (6) defines the binary user association variable.

Constraint (7) limits the transmit power of each beam and includes the possibility of beam shutdown when $p_{s,b}(t) = 0$.

Constraint (8) enforces the total transmit power budget of each satellite.

Constraint (9) ensures that the aggregate EPFD at the GS does not exceed the applicable EPFD limit.

The objective in (4) maximizes the weighted satisfaction of users. The weight $w_{\pi(u)}$ reflects the service priority of user u , while $s_u(t)$ represents the fraction of the user demand that is satisfied in time slot t . Therefore, higher-priority users can receive greater importance in the objective function.

The association and power allocation decisions are strongly coupled. User association affects beam loading, downlink rate, and user satisfaction, while beam power affects both downlink SINR and the aggregate EPFD at the GS. In near-inline NGSO–GSO geometries, shutting down beams can lower the aggregate EPFD, but it may also reduce the service rates of users originally served by the affected beams. Therefore, the problem becomes a mixed-integer nonlinear optimization problem. Solving the global optimum over all time slots is computationally difficult, so the next chapter develops EPFD-Aware Beam Reassociation for Service Recovery to obtain feasible association and power allocation decisions under the EPFD constraint.

IV. METHOD

A. Method Overview

Chapter III formulates the service recovery problem with user association, beam power, satellite power budget, and EPFD constraints. Under the EPFD constraint, beams from one or more LEO satellites may need to be shut down. A satellite is regarded as critical when at least one of its beams covers the GS region and must be shut down to satisfy the EPFD constraint. It should be noted that the critical satellite is a time-dependent operational role rather than a permanent satellite identity. A satellite is not predefined as critical before entering the near-inline period. It is regarded as critical only during the time slots in which its beams must be shut down due to the GS geometry and the EPFD constraint. After the satellite moves away from the GS direction and its closed beams return to normal operation, it is no longer treated as a critical satellite. Since this criterion is evaluated for all visible LEO satellites, multiple critical satellites may exist in the same time slot. To recover the service of users affected by beam shutdown on critical satellites, this chapter presents an EPFD-aware beam reassociation framework for service recovery.

The proposed framework is designed based on predictable satellite motion, beam footprints, beam-level EPFD contributions, and user service requirements. Instead of assuming a fixed critical-helper geometry, the framework determines the beam roles in each time slot according to the current satellite geometry and EPFD condition. The framework first identifies critical satellites, closed beams, critical safe beams, helper release beams, and helper recovery beams. It then constructs candidate beam relation graphs to describe the possible reassociation relations between critical beams and helper beams.

The proposed framework consists of three major online procedures. First, Safe-Beam Reassociation for Helper Users

(SBR) uses the residual capacity of critical safe beams to reassociate part of the helper users originally served by helper release beams. This reduces the loading of helper release beams and releases part of the helper satellite transmit power. Second, the released power is allocated to helper recovery beams according to their priority-weighted recovery demand. Third, Helper-Beam Reassociation for Closed-Beam Users (HBR) uses helper recovery beams to recover closed-beam users. Through these procedures, the proposed framework aims to improve service continuity for closed-beam users while maintaining EPFD compliance and power constraints.

In summary, the proposed EPFD-aware beam reassociation framework consists of the following steps:

- **Time-slot beam role record:** Precompute the critical satellites, closed beams, critical safe beams, helper release beam candidates, helper recovery beam candidates, and candidate beam relation graphs for each time slot.
- **Safe-Beam Reassociation for Helper Users (SBR):** Reassociate selected helper users from helper release beams to critical safe beams, reducing helper beam loading and releasing helper satellite transmit power.
- **Released power allocation for helper recovery beams:** Allocate the released transmit power to helper recovery beams according to the priority-weighted recovery demand.
- **Helper-Beam Reassociation for Closed-Beam Users (HBR):** Reassociate selected closed-beam users to helper recovery beams and recover their service under the available power of helper recovery beams.

B. Time-Slot Beam Role Record

Before the online reassociation procedures are executed, the system constructs a time-slot beam role record for each time slot. This record stores the critical satellites, helper satellites, and the candidate beam relations between critical beams and helper beams in the current time slot. These records are mainly built according to predictable satellite geometry, beam footprints, and beam-level EPFD contributions.

1) Critical Set Identification: For each time slot t , the system first checks whether the aggregate EPFD generated by all active beams of visible LEO satellites satisfies the EPFD limit. If the aggregate EPFD is already below the limit, beam shutdown is not required in this time slot, and the critical satellite set is empty. If the aggregate EPFD exceeds the limit, the system examines the beam-level EPFD contribution of each active beam on all visible LEO satellites and sorts the beams in descending order according to $\text{EPFD}_{s,b}(t)$. The system then sequentially shuts down beams with higher EPFD contributions and updates the aggregate EPFD after each beam shutdown until the aggregate EPFD satisfies the EPFD limit.

Let $\mathcal{B}_s^{\text{off}}(t)$ denote the set of beams shut down on satellite s in time slot t . After the beam shutdown process is completed, all satellites with a non-empty $\mathcal{B}_s^{\text{off}}(t)$ form the critical satellite set, denoted by $\mathcal{S}^C(t)$.

For each critical satellite s , the beams that are not shut down and remain active are denoted by $\mathcal{B}_s^{\text{safe}}(t)$, and are referred to as safe beams. Therefore, the critical satellite set is determined by the beam shutdown result of each time slot. If beams from

different satellites must be shut down to satisfy the EPFD constraint, these satellites are included in $\mathcal{S}^C(t)$ at the same time.

2) Helper Set Identification: After critical set identification is completed, the system constructs the helper satellite set, denoted by $\mathcal{S}^H(t)$. Similar to the critical satellite role, the helper satellite is also a time-dependent operational role rather than a permanent satellite identity. A neighboring satellite is regarded as a helper only when its beams have feasible overlap relations with the current critical satellites and can support the service recovery process in the current time slot. When the corresponding critical satellites leave the critical period and their beams return to normal operation, the helper relation is also released. For each helper satellite candidate, the system checks the footprint overlap between its beams and the safe beams and closed beams of critical satellites to determine the possible role of each helper beam.

If the service area of a helper beam overlaps with the footprint of critical safe beams, this helper beam is marked as a helper release beam candidate, denoted by $\mathcal{B}_h^{\text{ris}}(t)$. A helper release beam candidate represents a helper beam whose served users overlap with critical safe beams and whose loading may be reduced through reassociation.

If the service area of a helper beam overlaps with the footprint of critical closed beams, this helper beam is marked as a helper recovery beam candidate, denoted by $\mathcal{B}_h^{\text{rec}}(t)$. A helper recovery beam candidate represents a helper beam whose coverage can include users affected by closed beams.

If a helper satellite h has at least one helper release beam candidate or one helper recovery beam candidate, the satellite is included in the helper satellite set $\mathcal{S}^H(t)$.

3) Candidate Beam Relation Graph Construction: After the critical set and helper set are identified, the system constructs candidate beam relation graphs to record beam-level overlap relations. These graphs indicate which beam pairs have footprint overlap and therefore form candidate reassociation relations.

As shown in Fig. 2, the system first constructs the safe-release candidate graph. One side of this graph consists of critical safe beams, and the other side consists of helper release beam candidates. If the footprint of a critical safe beam overlaps with the service area of a helper release beam, an edge is created between them. Each edge represents a safe-release candidate relation and records the overlap between a critical safe beam and a helper release beam.

As shown in Fig. 3, the system next constructs the closed-recovery candidate graph. One side of this graph consists of critical closed beams, and the other side consists of helper recovery beam candidates. If the footprint of a helper recovery beam overlaps with the service area of a closed beam, an edge is created between them. Each edge represents a closed-recovery candidate relation and records the overlap between a critical closed beam and a helper recovery beam.

Through these graph constructions, the system organizes the possible beam-level relations between critical beams and helper beams before the online reassociation procedures are executed.

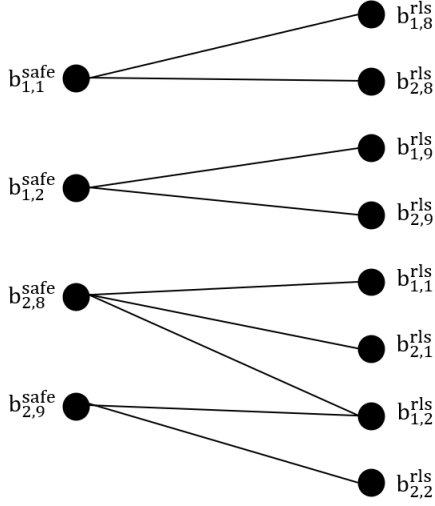


Fig. 2. Safe-release candidate graph between critical safe beams and helper release beam candidates. The left-side nodes $b_{s,\ell}^{safe}$ denote safe beams on critical satellites, and the right-side nodes $b_{h,k}^{rls}$ denote helper release beam candidates. An edge $e = (b_{s,\ell}^{safe}, b_{h,k}^{rls})$ exists when the two beams have footprint overlap.

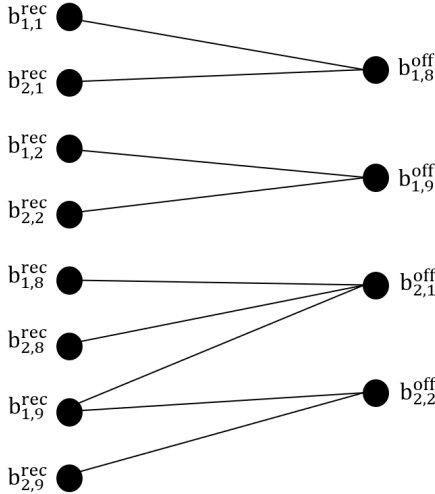


Fig. 3. Closed-recovery candidate graph between critical closed beams and helper recovery beam candidates. The left-side nodes $b_{s,\ell}^{off}$ denote closed beams on critical satellites, and the right-side nodes $b_{h,k}^{rec}$ denote helper recovery beam candidates. An edge $e = (b_{s,\ell}^{off}, b_{h,k}^{rec})$ exists when the two beams have footprint overlap.

It should be noted that the candidate beam relation graphs are used to record feasible reassociation relations, rather than to directly formulate a graph matching problem. An edge in these graphs only indicates that two beams have footprint overlap and may form a candidate reassociation relation. The edge score is not fixed when the graph is constructed. Instead, the actual release score or recovery score is evaluated later according to the current user subset, residual beam capacity, helper beam loading, available helper power, and user satisfaction. After one edge is selected, the user association, remaining capacity, and available helper resources are updated, so the scores of the remaining edges may also change. Therefore, the

graph record provides the feasible candidate relations for SBR and HBR, while the final active relations are determined by the following score evaluation and iterative selection procedures.

C. Safe-Beam Reassociation for Helper Users

After the time-slot beam role record is completed, the system performs Safe-Beam Reassociation for Helper Users (SBR). The purpose of SBR is to use the residual capacity of critical safe beams to serve part of the users originally served by helper release beams. After these helper users are reassociated to critical safe beams, the loading of the corresponding helper release beams is reduced, allowing the helper satellite to release part of its transmit power.

SBR takes the safe-release candidate graph constructed in the time-slot beam role record as its input. Based on this graph, SBR evaluates the candidate helper users on each safe-release edge and determines the active safe-release relations.

For each edge in the safe-release candidate graph, SBR first identifies the helper users located in the overlap region between the critical safe beam and the helper release beam. Let $\mathcal{U}_e^H(t)$ denote the candidate helper users covered by edge e . These users are originally served by the helper release beam and are located in the overlap region corresponding to the edge. In addition, only helper users that can maintain their service satisfaction after being reassociated to the critical safe beam are included in $\mathcal{U}_e^H(t)$.

Next, SBR calculates how much helper load can be reassociated through the edge according to the residual capacity of the critical safe beam. Let L_u denote the service load of user u , and let $C_e^{safe}(t)$ denote the residual capacity of the critical safe beam corresponding to edge e after serving its original users. SBR needs to select a feasible subset of helper users from $\mathcal{U}_e^H(t)$, such that the total load of the selected users does not exceed $C_e^{safe}(t)$ and is as large as possible. Therefore, the release score of edge e is defined as follows:

$$\text{Score}_e^{\text{SBR}}(t) = \max_{\mathcal{A} \subseteq \mathcal{U}_e^H(t)} \sum_{u \in \mathcal{A}} L_u \quad (4)$$

subject to

$$\sum_{u \in \mathcal{A}} L_u \leq C_e^{safe}(t), \quad (5)$$

where $\mathcal{A}$ denotes a reassociation user subset selected from the candidate helper users. The user subset that achieves this maximum value is denoted by $\mathcal{A}_e^{\text{SBR}}(t)$. Therefore, $\text{Score}_e^{\text{SBR}}(t)$ represents the maximum helper service load that can be reassociated by activating edge e under the current residual capacity of the safe beam. In other words, SBR does not only consider the total helper load on the edge. It further checks which users can actually fit within the residual capacity of the safe beam and how much load can be reassociated.

To select active safe-release relations, SBR performs an iterative greedy activation procedure on the safe-release candidate graph. In each iteration, the system recalculates the release scores of the unselected edges according to the current graph state. The edges that have already been activated are recorded in the active safe-release relation set and are marked as selected in later iterations, so they no longer participate

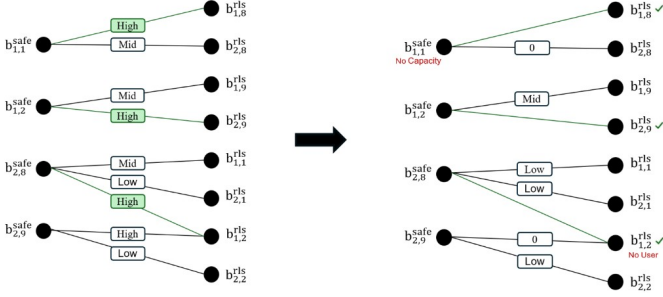


Fig. 4. Example of iterative edge activation and graph update in SBR.

in the next selection. For the currently available edges, the system determines connected components based on whether the edges share critical safe beams or helper release beams. Different connected components do not share the related beam resources and can therefore be processed independently.

For each connected component, SBR selects the edge with the highest release score greater than zero for activation. If multiple edges have the same highest release score, the system randomly selects one of them. For each selected edge, SBR reassociates the helper users in $\mathcal{A}_e^{\text{SBR}}(t)$ to the corresponding critical safe beam. After the reassociation is completed, the edge is recorded as an active safe-release relation.

As shown in Fig. 4, after each iteration, SBR records the edges activated in the current iteration into the active safe-release relation set and updates the original safe-release candidate graph. The updated information includes the user association state, the residual capacity of critical safe beams, the loading of helper release beams, and the released transmit power of helper satellites. The system then recalculates the release scores of the remaining unselected edges according to the updated graph state.

This iterative process continues until the release scores of all unselected edges in the safe-release candidate graph become zero. After SBR is completed, the system obtains the active safe-release relation set, the updated user association state, and the transmit power released by each helper satellite.

D. Released Power Allocation for Helper Recovery Beams

After SBR is completed, the transmit power released by helper release beams is added back to the available power budget of the corresponding helper satellite. The system then allocates this available power to helper recovery beams on the same helper satellite as the available power of helper recovery beams for serving closed-beam users.

To determine the power allocation order, the system first checks the candidate users connected to each helper recovery beam according to the closed-recovery candidate graph. These candidate users come from critical closed beams that have footprint overlap with the helper recovery beam. The system then calculates the priority-weighted recovery demand of each helper recovery beam based on the service priorities and traffic demands of these candidate users. If a helper recovery beam is connected to more high-priority users or users with larger traffic demand, the beam has a higher recovery demand value.

After the recovery demand value is calculated, the helper recovery beams are sorted in descending order of priority-weighted recovery demand. The released power is then allocated to the helper recovery beams according to this order. Each helper recovery beam can be increased at most to its beam power upper bound. If a beam reaches its upper bound, the remaining released power is allocated to the next helper recovery beam. This process continues until the released power is exhausted or all helper recovery beams have reached their beam power upper bounds.

E. Helper-Beam Reassociation for Closed-Beam Users

After released power allocation is completed, the system performs Helper-Beam Reassociation for Closed-Beam Users (HBR) to recover the service of users located under critical closed beams. The purpose of HBR is to use the available power of helper recovery beams to reassociate users affected by critical beam shutdown to available helper recovery beams.

HBR takes the closed-recovery candidate graph constructed in the time-slot beam role record as its input. Based on this graph, HBR evaluates the candidate closed-beam users on each closed-recovery edge and determines the active closed-recovery relations.

For each edge in the closed-recovery candidate graph, HBR first identifies the closed-beam users located in the overlap region between the critical closed beam and the helper recovery beam. Let $\mathcal{U}_e^C(t)$ denote the candidate closed-beam users covered by edge e . These users are originally located in the service area of a critical closed beam and need to be reassociated to another available beam due to beam shutdown.

Next, HBR determines which closed-beam users can be served by the edge according to the available power of the helper recovery beam. Let $P_e^{\text{avail}}(t)$ denote the available power of the helper recovery beam corresponding to edge e . The users in $\mathcal{U}_e^C(t)$ are first sorted by service priority, where higher-priority users are considered before lower-priority users. Within the same priority level, users with better resource efficiency are considered first, meaning that users requiring less power to achieve the required service satisfaction are served earlier. HBR then sequentially admits users according to this ordering under $P_e^{\text{avail}}(t)$.

Let $\mathcal{A}_e^{\text{HBR}}(t) \subseteq \mathcal{U}_e^C(t)$ denote the set of closed-beam users selected by this priority-ordered sequential admission process. If the remaining power is insufficient to fully satisfy the next ordered user, the user can still be included in $\mathcal{A}_e^{\text{HBR}}(t)$ with partial satisfaction. Therefore, $\mathcal{A}_e^{\text{HBR}}(t)$ contains the users that are reassociated through edge e under the current available power of the helper recovery beam.

Based on $\mathcal{A}_e^{\text{HBR}}(t)$, HBR calculates the recovery score of the edge. Let $w_{\pi(u)}$ denote the weight of the service priority level of user u , and let $s_{u,e}(t)$ denote the satisfaction that user u can achieve after being reassociated through edge e . The recovery score of edge e is defined as follows:

$$\text{Score}_e^{\text{HBR}}(t) = \sum_{u \in \mathcal{A}_e^{\text{HBR}}(t)} w_{\pi(u)} \cdot s_{u,e}(t). \quad (6)$$

This recovery score represents the priority-weighted service recovery benefit that can be achieved by activating edge e .

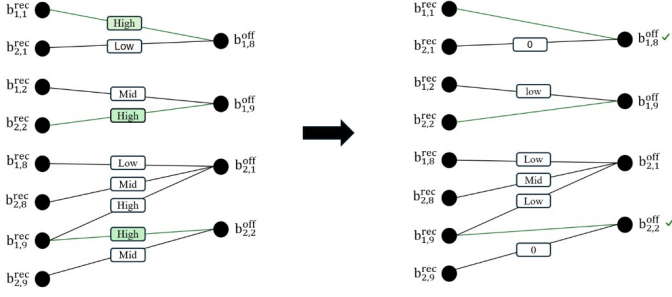


Fig. 5. Example of iterative edge activation and graph update in HBR.

under the current available power of the helper recovery beam. If an edge can recover more high-priority users or provide higher weighted satisfaction under the same available power, the edge obtains a higher recovery score.

To select active closed-recovery relations, HBR performs an iterative greedy activation procedure on the closed-recovery candidate graph. In each iteration, the system recalculates the recovery scores of the unselected edges according to the current graph state. The edges that have already been activated are recorded in the active closed-recovery relation set and are marked as selected in later iterations, so they no longer participate in the next selection. For the currently available edges, the system determines connected components based on whether the edges share critical closed beams or helper recovery beams. Different connected components do not share the related beam resources and can therefore be processed independently.

For each connected component, HBR selects the edge with the highest recovery score greater than zero for activation. If multiple edges have the same highest recovery score, the system randomly selects one of them. For each selected edge, HBR reassociates the closed-beam users in $\mathcal{A}_e^{\text{HBR}}(t)$ to the corresponding helper recovery beam. After the reassociation is completed, the edge is recorded as an active closed-recovery relation.

As shown in Fig. 5, after each iteration, HBR records the edges activated in the current iteration into the active closed-recovery relation set and updates the original closed-recovery candidate graph. The updated information includes the user association state, the loading of helper recovery beams, the available power of helper recovery beams, and the closed-beam users that have not yet been recovered. The system then recalculates the recovery scores of the remaining unselected edges according to the updated graph state.

This iterative process continues until the recovery scores of all unselected edges in the closed-recovery candidate graph become zero. After HBR is completed, the system obtains the active closed-recovery relation set, the updated user association state, and the service recovery result of closed-beam users.

F. Overall Procedure and State Update

This section summarizes the overall procedure of EPFD-Aware Beam Reassociation for Service Recovery. The previous subsections describe the time-slot beam role record, SBR,

released power allocation, and HBR. This section connects these steps into a complete time-slot procedure and explains the system state that must be updated after each time slot.

Before the online reassociation procedures are executed, the system precomputes the time-slot beam role records for all time slots according to predictable satellite motion, beam footprints, user coverage, and beam-level EPFD contributions. If the aggregate EPFD of a time slot already satisfies the EPFD limit, beam shutdown is not required in that time slot. If the aggregate EPFD exceeds the limit, the corresponding critical satellites, closed beams, critical safe beams, helper release beam candidates, and helper recovery beam candidates are stored in the record.

After the beam role records are completed, the system further constructs the safe-release candidate graphs and the closed-recovery candidate graphs. The safe-release candidate graph is used as the input of SBR to determine which helper users may be reassociated to critical safe beams, thereby releasing helper satellite transmit power. The closed-recovery candidate graph is used as the input of HBR to determine which closed-beam users may be reassociated to helper recovery beams.

In each time slot, the system retrieves the corresponding time-slot beam role record. If no closed beam exists in the time slot, the system does not need to execute the following reassociation procedures. If closed beams exist, the system first performs SBR. SBR selects active safe-release relations according to the release score of each safe-release edge and reassociates selected helper users to the corresponding critical safe beams. After SBR is completed, the loading of helper release beams is reduced, and the corresponding released transmit power is added to the available power budget of the helper satellite.

Before HBR is executed, the system performs released power allocation. This step allocates the available power of the helper satellite to helper recovery beams according to their priority-weighted recovery demand. After the allocation, each helper recovery beam obtains an updated available power.

Finally, the system performs HBR. HBR selects active closed-recovery relations according to the recovery score of each closed-recovery edge and reassociates selected closed-beam users to the corresponding helper recovery beams.

Algorithm 1 Overall Procedure of EPFD-Aware Beam Reassociation for Service Recovery

Input: LEO constellation and beam configuration, GSO–GS and EPFD model, user distribution with service demand and priority weights, and beam power configuration

Output: User association sequence $\mathbf{X}$ and beam power allocation sequence $\mathbf{P}$

```

1: Initialize  $\mathbf{X}(t)$  and  $\mathbf{P}(t)$ 
2: Precompute the time-slot beam role records for all time slots
3: Construct the safe-release candidate graphs for all time slots
4: Construct the closed-recovery candidate graphs for all time slots
5: for each time slot  $t$  do
6:   Retrieve the time-slot beam role record at time slot  $t$ 
7:   if no closed beam exists then
8:     continue
9:   end if
10:  // Safe-Beam Reassociation for Helper Users
11:  Initialize the active safe-release relation set as empty
12:  Evaluate the release score of each safe-release edge by (4)
13:  while there exists a safe-release edge with a positive release score do
14:    Divide the current safe-release candidate graph into connected components
15:    for each connected component do
16:      Select the edge with the maximum release score
17:      Reassociate the helper users in  $\mathcal{A}_e^{\text{SBR}}(t)$  to the corresponding critical safe beam
18:      Add this edge to the active safe-release relation set
19:      Mark this edge as selected
20:    end for
21:    Update the safe-release candidate graph and recompute the release scores by (4)
22:  end while
23:  // Released Power Allocation for Helper Recovery Beams
24:  Compute the priority-weighted recovery demand of each helper recovery beam
25:  Sort helper recovery beams in descending order of recovery demand
26:  Allocate the released power to helper recovery beams subject to beam and satellite power limits
27:  Update the available power of helper recovery beams
28:  // Helper-Beam Reassociation for Closed-Beam Users
29:  Initialize the active closed-recovery relation set as empty
30:  Evaluate the recovery score of each closed-recovery edge by (6)
31:  while there exists a closed-recovery edge with a positive recovery score do
32:    Divide the current closed-recovery candidate graph into connected components
33:    for each connected component do
34:      Select the edge with the maximum recovery score
35:      Reassociate the closed-beam users in  $\mathcal{A}_e^{\text{HBR}}(t)$  to the corresponding helper recovery beam
36:      Add this edge to the active closed-recovery relation set
37:      Mark this edge as selected
38:    end for
39:    Update the closed-recovery candidate graph and recompute the recovery scores by (6)
40:  end while
41:  Update  $\mathbf{X}(t)$  and  $\mathbf{P}(t)$ 
42: end for
43: return  $\mathbf{X}, \mathbf{P}$ 

```

V. DISCUSSION

A. Applicability to Different Multi-Beam LEO Constellations

The proposed EABR framework is designed for general multi-beam LEO satellite systems and is not limited to a specific constellation. Although the evaluation in this work adopts a OneWeb-like constellation as the simulation case, the core idea of EABR depends on predictable satellite motion, beam footprints, beam-level EPFD contributions, and coverage overlap among neighboring LEO satellites. These properties are common in many multi-beam LEO systems.

Therefore, as long as the satellite geometry and beam coverage can be estimated in advance, the time-slot beam role record and candidate beam relation graphs can be constructed for different constellation configurations. The proposed framework can then identify the critical satellites and helper satellites according to the current system state. In this sense, EABR is applicable to different multi-beam LEO constellations, while the OneWeb-like setting is used as a representative case for evaluation.

B. Helper Availability and Resource-Limited Recovery

The performance of helper-based service recovery depends on the availability of neighboring helper satellites and helper beams. In practical systems, helper availability is affected by constellation-level resource conditions, such as satellite density, orbital spacing, beam footprint overlap, EPFD limit strictness, and the remaining transmit power of neighboring satellites. Therefore, insufficient helper resources should be regarded as a system-level resource limitation rather than a failure of the reassociation framework.

When helper satellite resources are limited, the recovery performance of EABR is bounded by the available helper coverage, helper beam capacity, and helper satellite transmit power. Nevertheless, as long as partial helper coverage or helper power is available, EABR can still exploit these resources for service recovery and achieve higher user satisfaction than conventional interference mitigation approaches. The framework also maintains the EPFD constraint and power constraints during the reassociation process, ensuring that service recovery does not introduce additional interference violation.

VI. EVALUATION

A. Evaluation Overview

This chapter evaluates the performance of the proposed EPFD-Aware Beam Reassociation for Service Recovery (EABR) framework, focusing on its ability to improve user satisfaction and service continuity under the aggregate EPFD constraint. The evaluation consists of three main parts. First, we observe the aggregate EPFD variation during the critical period near the GS, as well as the number of beams that must be shut down to satisfy the EPFD constraint. This analysis confirms the existence of EPFD violation and beam shutdown demand during the critical period, and further supports the rationale of designing a reassociation-based service recovery framework. Second, we compare different interference mitigation methods in terms of service performance and service recovery capability. Finally, we further analyze the impact of different EPFD limits on user satisfaction to evaluate the stability and applicability of EABR under different interference constraint conditions.

B. Simulation Setup

To evaluate the performance of the proposed EABR framework under the NGSO–GSO coexistence scenario, we design a simulation framework that combines STK and MATLAB. This framework is used to construct the LEO constellation, obtain the time-varying satellite geometry, and further evaluate the performance of different interference mitigation methods in terms of EPFD compliance, downlink capacity, and user satisfaction.

The simulation uses the following two main tools:

- **STK (Satellite Tool Kit):** [25] STK is used to construct the OneWeb-like LEO constellation and generate the relative positions, visibility relationships, and beam coverage information among the LEO satellites, the GS, and users at

TABLE I
SIMULATION PARAMETERS.

Parameter	Value
LEO constellation	
Altitude (km)	1200
Orbit plane inclination	87.9°
Time slot	1 s
Maximum transmit power	16.8 W per satellite
Maximum antenna gain	34.33 dBi
Downlink frequency	11.7 GHz
Bandwidth	250 MHz
Antenna pattern	[19] $A \exp(\beta\phi)$
Maximum tilt angle	10°
Approximated antenna gain coefficient, (A)	4.0×10^3
Approximated antenna gain coefficient, (B)	-0.0671
GSO satellite	
Altitude (km)	35785.4
Longitude	30.6
Latitude	0
Downlink frequency	11.7 GHz
GSO ground station	
Longitude	30.6
Latitude	0
Minimum elevation angle	10°
Maximum receive gain	38.1 dBi
Noise temperature	240 K
Antenna pattern	[12]
LEO user ground station	
Maximum receive gain	40.95 dBi
Noise temperature	240 K
Antenna pattern	[12]
User demand	50 Mbps
EPFD	
EPFD limit (dB(W/m ² /1 MHz))	-173.4 [9]
Reference antenna diameter	60 cm

each time slot. Based on the system information generated by STK, the dynamic geometric variation during the critical period near the GS can be simulated.

- **MATLAB:** [26] MATLAB is used to process the system information generated by STK and perform offline preprocessing. This includes constructing the beam-user association, critical/helper satellite relationships, time-slot beam role records, and candidate beam relation graphs. MATLAB is then used to execute the proposed EABR framework and calculate the EPFD and user satisfaction performance of each method at different time slots.

The main simulation parameters used in this chapter are summarized in Table I. The table integrates the settings related to the LEO constellation, GSO satellite, GS, LEO user ground station, and EPFD evaluation. These parameters are used in the following EPFD evaluation and user satisfaction comparison.

C. Compared Schemes

To evaluate the proposed EABR framework, we compare it with several interference mitigation methods. All methods are evaluated under the same simulation parameters to ensure a fair comparison. The compared schemes are described as follows:

- **Beam shutdown only:** When the aggregate EPFD exceeds the limit, the high-interference beams on critical satellites are directly shut down to satisfy the EPFD constraint. This method does not perform beam reassociation for service recovery.

- **PC + Tilt:** This method represents a power-control and satellite-tilting based interference mitigation strategy. It reduces the interference toward the GS by adjusting the transmit power and satellite tilt angle. Although this method can help satisfy the EPFD constraint, it does not provide a reassociation-based service recovery mechanism for users.
- **Only HBR:** This method only applies HBR. Helper recovery beams serve users located under critical closed beams using the default power of the helper satellite. SBR and released power allocation are not applied.
- **EABR (Proposed):** This method includes SBR, released power allocation for helper recovery beams, and HBR.

D. Evaluation Results

Based on the simulation setup and compared schemes described above, this section evaluates the performance of EABR during the critical period. The analysis is divided into four parts. First, the aggregate EPFD and the number of closed beams are examined to verify whether EABR can still satisfy the EPFD constraint after reassociation-based service recovery, and to observe the beam shutdown behavior of critical satellites at different time slots. Second, the average satisfaction of different methods is compared under different user loads to analyze how increasing system load affects service performance. Third, the number of recovered closed-beam users is used to explain the main reason why EABR improves satisfaction, and the satisfaction CDF is further used to observe the satisfaction distribution of all users. Finally, the

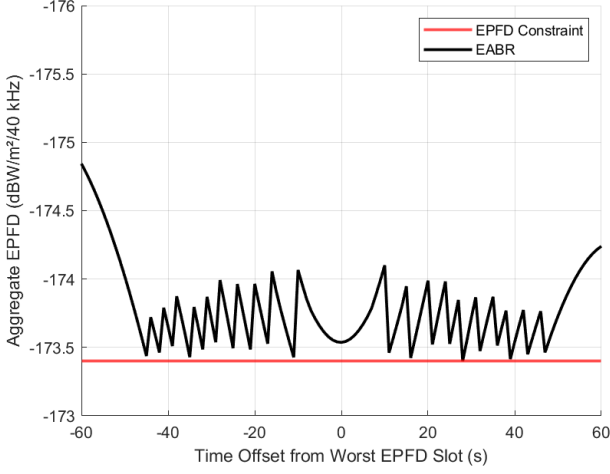


Fig. 6. Aggregate EPFD over time during the critical period.

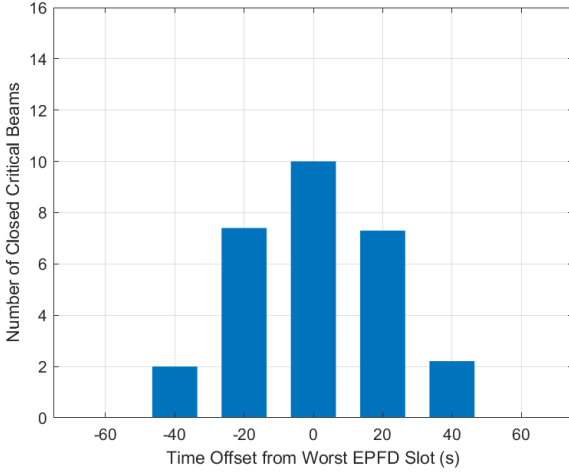


Fig. 7. Number of closed beams on critical satellites over time during the critical period.

average user satisfaction of EABR under different EPFD limits is evaluated to observe how the strictness of the interference limit affects service performance.

1) EPFD Compliance and Beam Shutdown:

This subsection first analyzes the aggregate EPFD after applying EABR and further observes the number of beams that must be shut down on critical satellites to satisfy the EPFD constraint. This analysis is used to verify whether the proposed framework can still satisfy the EPFD limit after reassociation-based service recovery, and to show the beam shutdown behavior during the critical period.

Fig. 6 shows the aggregate EPFD variation during the critical period near the GS. The red line represents the EPFD constraint, and the black line represents the aggregate EPFD after applying EABR. As shown in the figure, EABR keeps the aggregate EPFD below the limit during the entire observation period. This indicates that the proposed framework does not cause EPFD violation while performing service recovery for closed-beam users. Therefore, EABR can compensate for the

service loss caused by beam shutdown while maintaining EPFD compliance.

Fig. 7 shows the number of beams that must be shut down on critical satellites at different time offsets. When critical satellites approach the worst EPFD slot, the number of closed beams gradually increases and reaches its maximum around the worst EPFD slot. After the satellites move away from the GS, the number of closed beams gradually decreases. This result shows that the beam shutdown demand changes with the satellite-GS geometry.

2) User Satisfaction under Different User Loads:

This subsection compares the average satisfaction of different interference mitigation methods under different user loads. By changing the number of users, we can observe how the service performance changes as the system load increases, and evaluate whether EABR can maintain user satisfaction under different load conditions.

Fig. 8 shows the average satisfaction under three user load scenarios: 30 users, 50 users, and 70 users. As shown in the figure, Beam shutdown only has the lowest satisfaction because it does not provide service recovery for users affected by closed beams. PC + Tilt can reduce the interference toward the GS, but it does not recover the service of users affected by beam shutdown. Only HBR can serve part of the closed-beam users through helper recovery beams, so it achieves better satisfaction than Beam shutdown only and PC + Tilt.

Compared with the other methods, EABR maintains higher average satisfaction under all three user loads. When the user load is low, the helper satellite usually has enough available power, so the difference between Only HBR and EABR is small. However, as the user load increases, more closed-beam users need service recovery, and Only HBR becomes limited by the default power of the helper satellite. In contrast, EABR uses SBR to release available power from the helper satellite and improve the service recovery capability of helper recovery beams. Therefore, EABR can maintain more stable service performance under medium and high user loads.

3) Service Recovery Capability and Satisfaction Distribution:

This subsection further analyzes why EABR improves user satisfaction and observes the satisfaction distribution of different methods. Since 30 users is a low-load scenario, the default available power of the helper satellite is usually sufficient to support most recovery demand. Therefore, the difference in service recovery capability is not obvious under 30 users. To better show the effect of the proposed framework, this subsection focuses on the 50-user and 70-user scenarios.

Fig. 9 compares the number of successfully recovered closed-beam users between Only HBR and EABR under the 50-user and 70-user scenarios. As shown in the figure, EABR serves more closed-beam users in both load scenarios. The difference becomes more obvious around the worst EPFD slot, where the beam shutdown demand is the most severe. This result shows the necessity of SBR in EABR.

Fig. 10 shows the satisfaction CDF of three interference mitigation methods. This figure focuses on the user-level satisfaction distribution. A CDF curve that shifts further to the right means that more users maintain higher satisfaction.

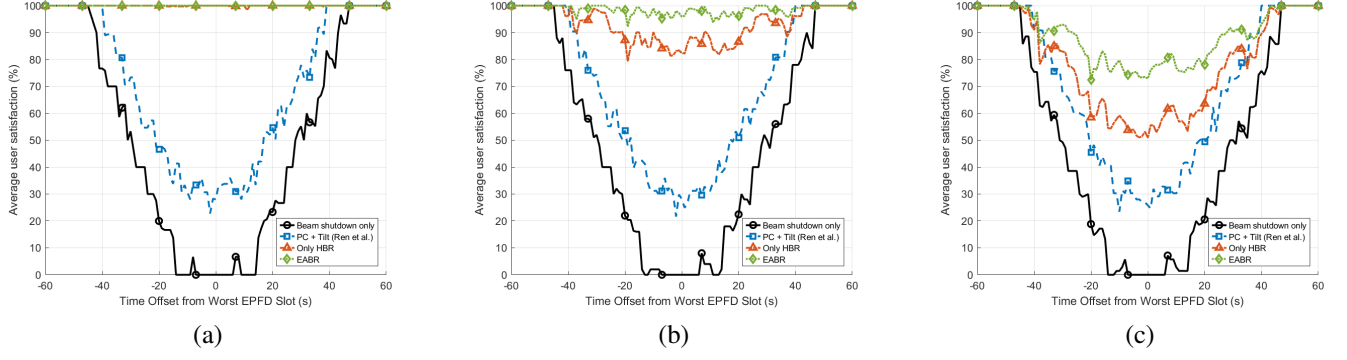


Fig. 8. Average user satisfaction of different interference mitigation methods under different user loads. (a) 30 users. (b) 50 users. (c) 70 users.

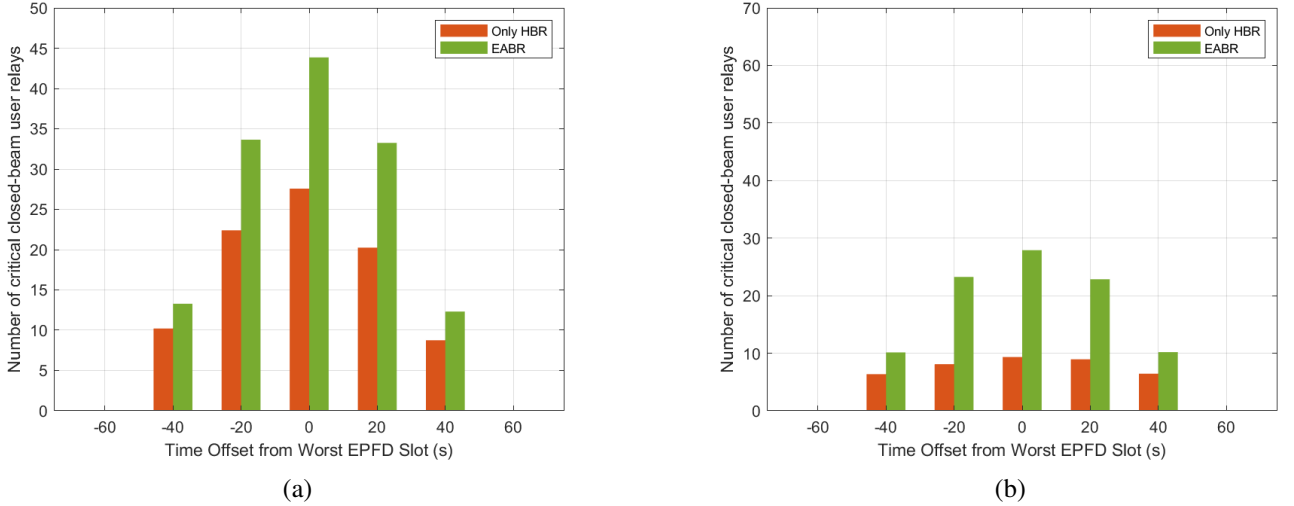


Fig. 9. Number of recovered closed-beam users with and without SBR under different user loads. (a) 50 users. (b) 70 users.

In contrast, if the curve rises quickly in the low-satisfaction region, more users suffer from service degradation.

As shown in the figure, the CDF curve of EABR shifts more toward the high-satisfaction region than the other methods. This means that more users can maintain better service quality. The result shows that EABR not only improves average satisfaction, but also reduces the proportion of low-satisfaction users, making the overall user service quality more stable.

4) Impact of Different EPFD Limits:

This subsection further evaluates the impact of different EPFD limits on the service performance of EABR. When the EPFD limit becomes stricter, the allowed aggregate interference becomes lower. As a result, critical satellites may need to shut down more beams to satisfy the constraint, and the recovery demand of closed-beam users may also increase. This analysis is used to observe whether EABR can maintain stable average user satisfaction under different EPFD limits.

Fig. 11 shows the average user satisfaction of EABR under different EPFD limits. As shown in the figure, when the EPFD limit is more relaxed, EABR can maintain higher average user satisfaction. This result shows that the strictness of the EPFD limit directly affects user satisfaction. It also indicates that EABR can achieve better service performance under more

relaxed interference constraints.

Overall, EABR keeps the aggregate EPFD below the limit during the critical period and uses SBR to improve the ability of helper recovery beams to serve closed-beam users. Compared with other interference mitigation methods, EABR maintains higher average satisfaction under different user loads and reduces the proportion of low-satisfaction users. In addition, the results under different EPFD limits show that the strictness of the interference constraint affects user satisfaction, and EABR can maintain better service performance under more relaxed constraints. Overall, EABR achieves a better balance between EPFD compliance and service continuity.

VII. CONCLUSION

This thesis proposed EPFD-Aware Beam Reassociation for Service Recovery (EABR), a beam reassociation framework for EPFD-constrained multi-beam LEO satellite systems under NGSO–GSO coexistence. EABR exploits predictable satellite motion, beam footprints, and coverage overlap among neighboring LEO satellites to support users affected by critical-beam shutdown. The framework consists of Safe-Beam Reassociation for Helper Users (SBR), released power allocation for helper recovery beams, and Helper-Beam Reassociation for Closed-Beam Users (HBR).

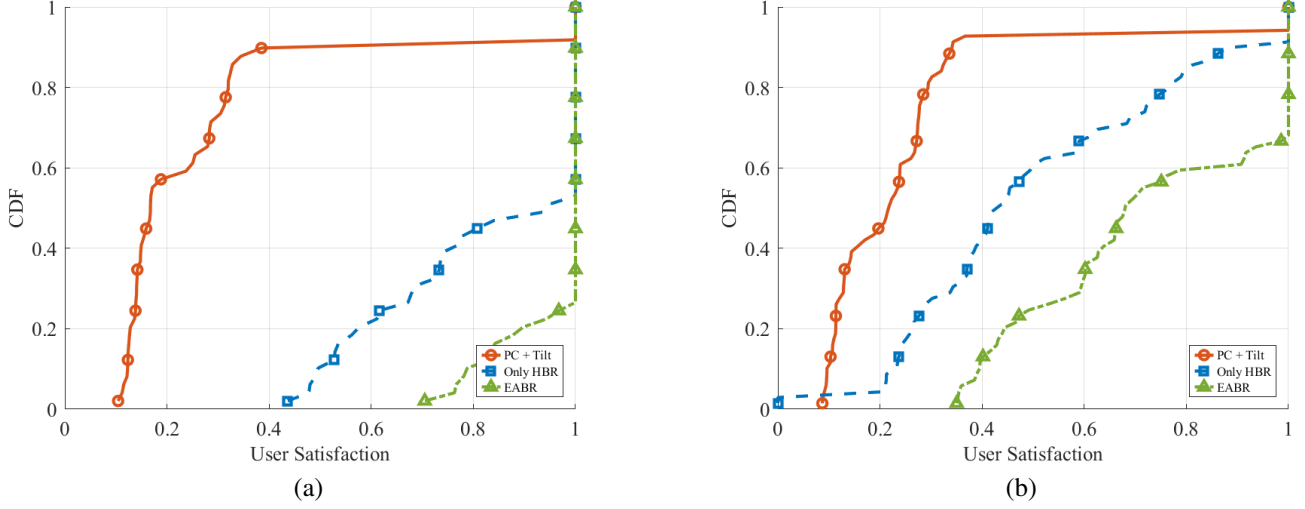


Fig. 10. CDF of user satisfaction for different interference mitigation methods under different user loads. (a) 50 users. (b) 70 users.

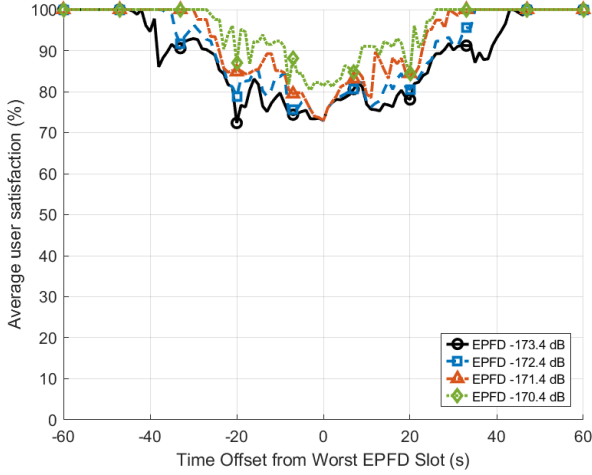


Fig. 11. Average user satisfaction of EABR under different EPFD limits.

The simulation results show that EABR keeps the aggregate EPFD below the required limit during the critical period. Compared with beam shutdown only, PC + Tilt, and Only HBR, EABR achieves higher average user satisfaction under different user loads. The results also show that SBR increases the number of recovered closed-beam users, especially around the worst EPFD slot, and the satisfaction CDF confirms that EABR reduces the proportion of low-satisfaction users.

Overall, EABR provides a practical service recovery approach for multi-beam LEO satellite systems under the GS EPFD constraint. By using precomputed beam relation information and neighboring helper satellites, the proposed framework improves service continuity without relying on high-complexity global optimization in every time slot. Therefore, EABR achieves a better balance between EPFD compliance, user satisfaction, and implementation complexity.

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